The Mathematical Model of Induction Heating of Ferromagnetic Pipes

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Abstract—**In this paper a mathematical model of coupled heat transfer and electromagnetic phenomena in induction heaters of ferromagnetic, steel pipes has been described. The model takes into account the nonlinearity of all coefficients, the characteristics of the supply source, and the thermal influence of the lining. A method of partial decoupling of fields has been proposed. Experimental verification of the computed results has also been presented.**

I. Introduction

OUR TASK is to build a mathematical model of physical phenomena which occur during a process of induction heating of ferromagnetic pipes. The phenomena are of electromagnetic, thermal, metallurgical, and strength types. The most characteristic feature of induction heating seems to be the fact that all of them are closely coupled. Because of the complexity, only the two dominating fields—eletromagnetic and thermal—are to be considered in this model. There are some factors which make the induction heaters difficult to design. The main factors are: the presence of several coupled physical fields and the need to describe them with partial differential equations, nonlinearity of all coefficients, complex heat exchange among all parts of the heater, great variety of geometric configurations and shapes of heated bodies, and a large number of possible manufacturing processes.

II. Mathematical Description

The coupled electromagnetic and temperature fields in an induction heater may be expressed by a set of differential equations—the Maxwell equations for the electromagnetic field and the Fourier–Kirchhoff equation for the temperature fields [3].

We assume that the pipe and the coil are both infinite and axisymmetrical, the bodies are idealized homogeneous and isotropic, the presence of coil windings is ignored when the heat transfer phenomena are considered, the heat generated due to hysteresis loop is neglected in comparison with that due to eddy currents.

When all the assumptions have been accepted, the two fields may be described in cylindrical coordinates in the following way:

$$\frac{\partial(\mu_2(H, T) * H(r, t))}{\partial t} - \frac{1}{r} * \frac{\partial}{\partial r}\left(r\rho_2(T) * \frac{\partial H(r, t)}{\partial r}\right) = 0 \quad (1)$$

$$\gamma_2(T)\, c_2(T)\, \frac{\partial T_2(r, t)}{\partial t} - \frac{1}{r} * \frac{\partial}{\partial r}\left(r\lambda_2(T)\, \frac{\partial T_2(r, t)}{\partial r}\right) = \rho_2(T)\left(\frac{\partial H(r, t)}{\partial r}\right)^2 \quad (2)$$

$$\gamma_L(T)\, c_L(T)\, \frac{\partial T_L(r, t)}{\partial t} - \frac{1}{r} * \frac{\partial}{\partial r}\left(r\lambda_L(T)\, \frac{\partial T_L(r, t)}{\partial r}\right) = 0 \quad (3)$$

where H is the field strength, T is temperature, t is time, μ is magnetic permeability, ρ is electrical resistivity, γ is mass density, c is the specific heat, λ is thermal conductivity, r is the variable radius, and subscript L refers to the lining, subscript 2 to the heated body.

The initial and the boundary conditions which mainly determine the accuracy of the results and the running time can be formulated as follows:

For the electromagnetic field:

- the initial conduction

$$H(r, 0) = H_P \quad (4)$$

- the boundary condition at the external surface of the pipe.

When at the external surface magnetic flux density is forced, the electrical circuit consisting of a source and a heater can be described by an expression of the following type:

$$u_1(t) = Ri_1(t) + L\frac{di_1(t)}{dt} + \frac{d\Psi_2(t)}{dt} \quad (5)$$

where u_1 is the voltage of the source, i_1 is the current of the coil, R is the total resistance of the heater, L is the inductance of the coil and the air gap, and Ψ_2 is the flux in the pipe.

Applying the Ampere and Faraday laws one can write

$$u_1(t) = \frac{Rl_1}{w} H_e(t) + \frac{Ll_1}{w} * \frac{\partial H_e(t)}{\partial t} + \frac{2\Pi r_2 w}{\gamma_2} * \frac{\partial H_e(t)}{\partial r} \quad (6)$$

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where w is the number of windings of the coil, l_1 is the length of the coil, and subscript e refers to the external surface.

Since the resistance and the inductance of the heater are not known at the begining of the computations, they must be estimated in a short iterative process applying the equations from the classical induction heating theory to obtain the initial values.

When at the external surface the field strength is forced, the condition can be simply written in the following form:

$$H_e(t) = \frac{i_1(t) * w}{l_1} \tag{7}$$

the boundary condition at the internal surface of the pipe:

$$\frac{r_1 \mu_0}{2} * \frac{\partial H_i(t)}{\partial t} = \frac{1}{\gamma_2} * \frac{\partial H_i(t)}{\partial r} \tag{8}$$

where subscript i refers to the internal surface.

If we consider the case of a solid cylinder, from the assumption of axial symmetry one can write

$$\frac{\partial H}{\partial r} = 0, \qquad \text{for } r = 0. \tag{9}$$

For the temperature field in the pipe:

- the initial condition

$$T_2(r, 0) = T_p \tag{10}$$

- the boundary conditions

$$-\lambda_2(T) * \frac{\partial T_2}{\partial t} = \epsilon_2(T)\left((T_e + 273)^4 - (T_{L_i} + 273)^4\right) + \alpha_2 * (T_e - T_{L_i}) \tag{11}$$

$$\lambda_2(T) * \frac{\partial T_2}{\partial t} = \alpha_1 * (T_i - T_a) \tag{12}$$

where ϵ is thermal emissivity, α is the convective heat transfer coefficient, T_a is the appropriate ambient temperature.

For the temperature field in the lining:

- the initial condition

$$T_L(r, 0) = T_{L_p} \tag{13}$$

- the boundary conditions

$$T_{L_e} = T_{00} \tag{14}$$

where T_{00} is the average temperature of the coil cooling water and is assumed to be 50°C.

$$-\lambda_L(T) * \frac{\partial T_L}{\partial t} = \epsilon_L(T)\left((T_e + 273)^4 - (T_{L_i} + 273)^4\right) + \alpha_L * (T_e - T_{L_i}). \tag{15}$$

III. Change of Coefficients During the Heating Cycle

All the preceding equations are strongly nonlinear due to the dependence of all their coefficients on the field quantities: namely, on the field strength and temperature. They were given in an analytical form and tabulated at the first entry into the program.

For steel the thermal coefficients take the form

$$\lambda_2(T) = \lambda_0 + \lambda_1 * \left(\frac{T}{1000}\right) - \frac{\lambda_2}{ch(\lambda_3 * (T - 975)/1000)}, \quad \epsilon = (1.5, \cdots, 2.5) \text{ percent} \tag{16}$$

$$c_2(T) = c_0 + c_1 * \left(\frac{T}{1000}\right) + c_2 * \exp\left(c_3 * |T - 768|\right), \quad \epsilon = (2.2, \cdots, 4.9) \text{ percent} \tag{17}$$

$$\gamma_2(T) = \gamma_{20°C} * \frac{1}{1 + 3 * \alpha_E(T - 20)}, \quad \epsilon = (0.6, \cdots, 2.0) \text{ percent} \tag{18}$$

$$\alpha_E(T) = \alpha_0 + \alpha_1 * \left(\frac{T}{100}\right) \frac{\alpha_2}{ch(\alpha_3 * (T - 905)/100)} \tag{19}$$

where ϵ is the maximum error of approximation within the temperature range from 0°C to 1200°C.

For chamotte lining, the thermal coefficients take the following form:

$$\lambda_L(T) = (0.697 + 0.00064 * T) * 1.1163 \frac{\text{W}}{\text{m °C}} \tag{20}$$

$$c_L(T) = (880 + 0.23 * T) * 4186.8 \frac{\text{J}}{\text{kg °C}} \tag{21}$$

$$\gamma_L = 1850 \frac{\text{kg}}{\text{m}^3}. \tag{22}$$

The magnetic permeability has been found to be

$$\mu(H, T) = \mu_0 \left(1 + \left(\frac{\mu}{\mu_0} - 1\right) * \Phi(T)\right) \tag{23}$$

where

$$\Phi(T) = \begin{cases} 1 - \left(\frac{T}{T_C}\right)^6, & T < T_C \\ 0, & T \geq T_C \end{cases} \tag{24}$$

where T_C is the temperature of the Curie transition.

Some reasonable doubts may arise when magnetic permeability is to be calculated. According to [3] for such strong fields as are applied in induction heating ($H > 10^4$ A/m) we can make use of an initial magnetization curve which is an average for all kinds of steels containing 0.22, $\cdots$, 0.99 percent of carbon. The errors which may appear from taking into account an average curve instead of the proper one for a certain kind of steel are less than 2.5 percent.

Electrical resistivity as a function of temperature has been described in this simple form

$$\rho_2(T) = \rho_{20°\mathrm{C}} * (1 - B * (T - 20)) \tag{25}$$

where B is the temperature coefficient of resistance.

The thermal coefficients α_1, α_2, α_L which are functions of many factors, i.e., temperature, geometric shapes, and state of surfaces, have been approximated.

IV. Numerical Methods

Almost all analyses based on widely used methods, i.e., FEM, integral equations, provide the engineer with a precise map of field quantities [4], [5]. An experienced man finds the information useful in correcting the design of an electromagnetic device to make its operation better and safer. But when one faces the problem of designing the supply source for that device, one requires the global quantities of the load. A further step is to be added—from the field distribution the global quantities must be derived. The easier and cheaper it is done the better. The problem becomes extremely severe when the load changes its properties during the work—induction heaters seem to be the best examples of that. In such cases, the finite difference method is likely to be suitable due to a relatively small number of points at which the field is calculated.

As the finite difference method has been chosen, the solution region in the pipe was taken to be either the entire pipe wall or a surface layer having a thickness of $4 * \Delta_e$, where Δ_e accounted for magnetic permeability obtained from magnetization characteristic at the surface field strength H_e. The introduction of the finite approximation involves a leading error of $O(\Delta t, \Delta r)^2$.

Only (11) and (15) which described heat exchange between the body and the lining were treated as nonlinear; for the temperature appeared in them in the fourth power and even small errors in temperature determination might have caused great errors in the heat flow calculations.

The program was run with 63 and 31 grid points and for the latter grid the computed temperature error was 2.3 percent as compared to the results obtained in the first case with a smaller Δr.

V. Averaging Method

The comparison of the equivalent time constants of thermal and electromagnetic processes shows that these differ considerably. Thus the solving of equations of both fields with the same time step seems very ineffective. To avoid these calculations of the thermal process, characterized by great inertia, a method from nonlinear mechanics has been applied; namely, the "averaging method" [6]. The main idea of this method is to split two motions into "a quick one" (in our case, electromagnetic) and "a slow one" (thermal). It is assumed that the quick field is periodic between two time points at which the thermal field is calculated. In (2) the average value of the internal heat sources is placed instead of the instantaneous value

$$w_{\mathrm{av}} = \rho_2 * \left(\frac{\partial H_{\mathrm{av}}}{\partial r}\right)^2 \tag{26}$$

where w_{av} is the average value of internal heat sources and H_{av} is the average value of field strength found to be

$$H_{\mathrm{av}} = \frac{1}{2 * \Pi} * \int_{2\Pi n}^{2\Pi(n+1)} \sqrt[2]{H^2(r, \Theta)\, d\Theta}, \quad n = 0, 1, 2 \cdots. \tag{27}$$

Even though the method presented above reduced the calculation time, the time remained relatively long. During the numerical experiments, it was found that there was no need to compute the magnetic field distribution in the pipe for every period of the magnetic field. No significant changes in temperature distribution have been observed when the new average value of internal heat sources was used in calculations when the temperature of the external surface of the heated body was increased by 5°C. This means that during the whole heating cycle up to 1000°C, the heat sources are to be changed 200 times instead of 3 times as it happens in the case of classical heat problems commonly applied to induction heating. Only when the temperature reaches the Curie transition, when the magnetic permeability begins to slope, due to high temperature, internal heat sources have been calculated more often in order not to overlook the Curie transition. Later, we again return to the previous algorithm of "sampling" the temperature field.

The proposed method of decoupling the fields is an approximate one and should be applied very cautiously depending on the thermal inertia of the system, but for engineering calculations it seems to be a very useful one.

VI. Influence of Power Supply Source

The character of the supply source which feeds any electromagnetic device, and the properties of its control system exert great influence on the heating process. This important feature is not usually taken into account when one sets up a mathematical model of any process. Let us discuss the constraints we face in induction heating.

When the heater is fed from an ac machine medium-frequency generator, the frequency of the supply voltage remains unchanged during the whole cycle of heating. But

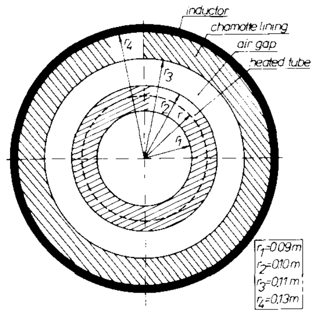

Fig. 1. Cross section of the considered induction heating system.

TABLE I
KEY VALUES OF THE HEATER

Supply voltage	3 × 380 V/50 Hz
Maximum ac output voltage of the converter	600 V
Maximum output current of dc link	250 A
Constant current of the inductor	426 A
Average frequency	2.2 kHz
Thyristor recovery time	80 μs
Length of coil	1.0 m
Number of windings	30 × ϕ20 Cu
Parallel capacitor	120 μF
Thermal constants:	$\epsilon_2 = 0.85$, $\epsilon_L = 0.8$, $\alpha_1 = 7\ \mathrm{W/m^2K}$, $\alpha_2 = 13\ \mathrm{W/m^2K}$, $\gamma_2(T)$; $\lambda_2(T)$; $c_2(T)$ same as for medium-carbon steel.

when we consider a thyristor as a parallel converter, as it has happened in our case, the output frequency can vary in a wide range—that change cannot be neglected at any rate. The parallel converter must work at a frequency which is a little higher than the frequency of self-oscillations of its resonant circuit. Also, it is varying when the heating cycle progresses due to the change in the inductance of the load when the capacitance remains unchanged.

The program contained a special loop that adjusted the frequency used in computations to the resonant frequency in such a way that the thyristor recovery time was at all times equal to that required.

It must also be determined which electrical quantity is controlled and stabilized by the source control system. Three main types are possible—when the supply power, the voltage across the inductor, or its current are fixed. We had the latter types, thus the boundary condition (7) was applied.

VII. Computation Results and Experimental Verification

In order to check the accuracy of the model outlined in this paper laboratory experiments have been carried out. An induction heater of the rate of 100 kW and output frequency of up to 3000 Hz was built (Fig. 1, Table I). In Figs. 2–4, the computed waveforms of the basic electromagnetic quantities are shown. In Fig. 2, the whole pipe possesses the ferromagnetic properties: in Fig. 3, the external layer of the pipe has already lost the ferromagnetic properties while the inner layer is still ferromagnetic. In Fig. 4, the whole material is nonmagnetic. In the first two stages the waveforms are highly distorted due to ferromagnetic properties.

A comparison between the results presented above and the ones obtained from a simple harmonic model with a

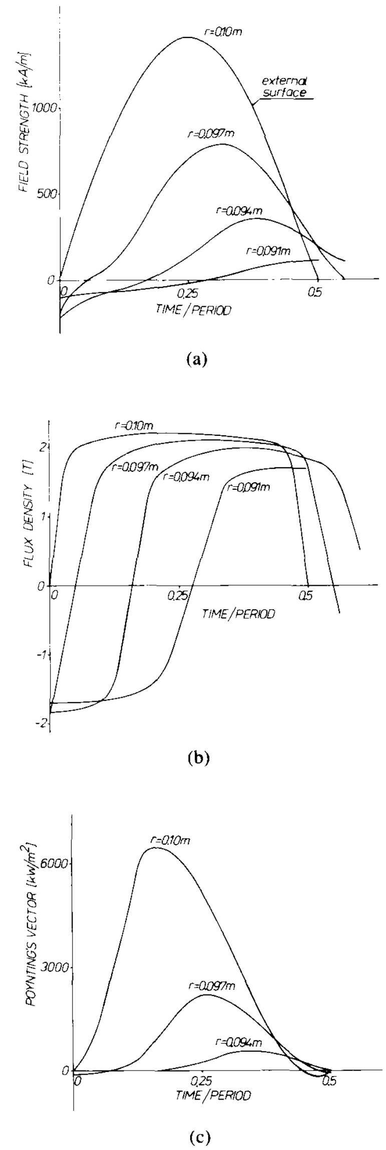

Fig. 2. Waveforms of basic electromagnetic quantities. Ferromagnetic state.

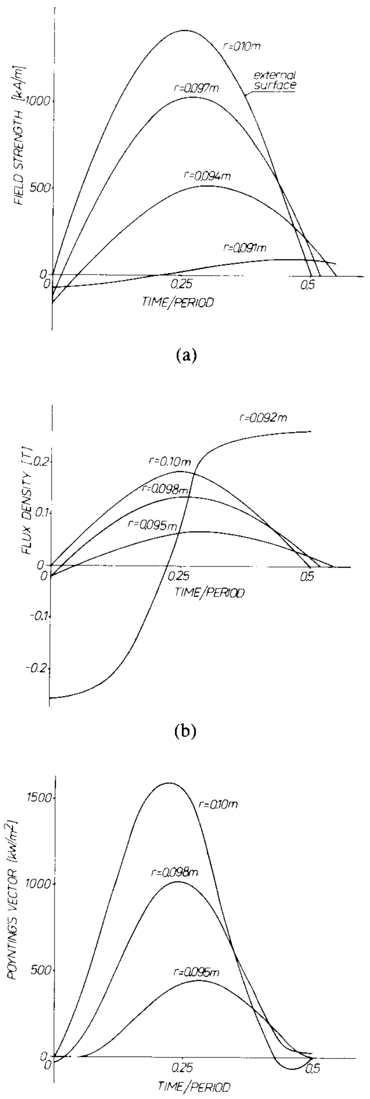

Fig. 3. Waveforms of basic electromagnetic quantities. Transient ferromagnetic state.

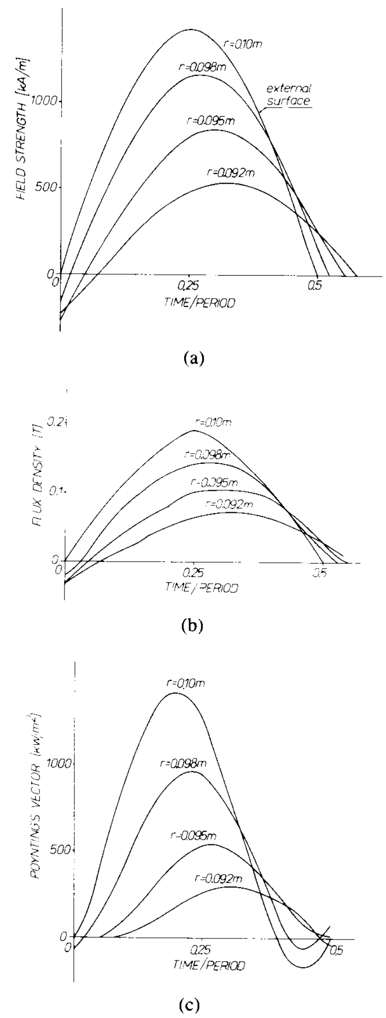

Fig. 4. Waveforms of basic electromagnetic quantities. Nonmagnetic state.

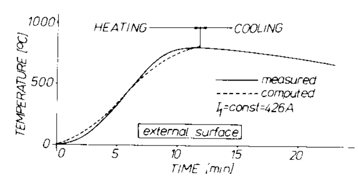

Fig. 5. Comparison between computed and measured temperature of the pipe.

magnetic permeability as a function of H, when the temperature distribution in the pipe was the same for both models, has been made [3], [7]. The differences between the two models have never been larger than 7 percent, which indicates that the influence of the distortions of electromagnetic quantities is in some way negligible when powers due to eddy currents are estimated.

The best way to check the accuracy of the model was to compare the computed and the measured temperatures of the pipe and the resistance (power) of the heater. One comparison is shown in Figs. 5 and 6. A close agreement is apparent and the obtained accuracy was better than 10 percent for temperature and 12 percent for resistance.

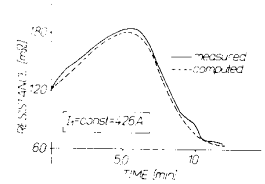

Fig. 6. Comparison between computed and measured resistance of the heater.

VIII. Conclusion

The unique attraction of the presented model is its mathematical simplicity, precisely written boundary conditions, taking into account the character of the supply source and the thermal influence of the lining. These features result in the utmost economy of computer storage and accuracy of computations. The proposed method of decoupling the fields has improved the economy of computations causing an insignificant decrease in accuracy, acceptable in most engineering problems, both in the field of electrical and process engineering.

So far, this model has been used by us in the process of time-optimal control of induction heating of steel pipes when a microprocessor system was applied, as a main controller.

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